

LISMORE, Australia

WMO: 945720

Lat: 28.8306S

Lon: 153.2600E

Elev: 9

StdP: 101.21

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	4.7	-2.1	3.2	15.8	-0.2	3.7	14.7	9.2	17.8	8.3	17.7	2.4	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	34.2	23.5	32.3	22.8	30.7	22.5	25.5	31.3	24.6	29.8	23.9	28.5	4.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	18.8	28.0	23.1	17.9	27.3	22.4	17.2	26.5	78.4	31.4	74.9	29.6	71.8	28.6	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.8	7.0	0.5	38.4	1.1	2.0	-0.3	39.8	-1.0	41.0	-1.6	42.1	-2.4	43.5		
			-0.1	26.6	1.4	1.1	-1.2	27.4	-2.0	28.0	-2.8	28.6	-3.8	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	24.1	23.8	22.5	19.8	16.4	14.4	13.6	14.6	17.4	19.6	21.6	23.1
	DBStd	4.35	2.21	2.15	1.84	1.75	2.17	2.25	2.18	2.28	2.50	2.62	2.56	2.46
	HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0
	HDD18.3	524	0	0	0	6	66	120	148	119	47	16	2	1
	CDD10.0	3364	438	386	387	293	199	131	112	142	222	298	349	405
	CDD18.3	844	180	153	129	49	7	1	1	3	19	56	101	147
	CDH23.3	6577	1464	1134	742	252	65	6	15	80	278	529	848	1165
	CDH26.7	2072	523	376	180	32	6	1	1	13	74	171	288	409
Wind	WSAvg	3.0	3.2	3.1	2.6	2.5	2.5	2.7	2.9	3.2	3.3	3.4	3.5	3.4
Precipitation	PrecAvg	1411	148	185	199	146	123	103	77	50	39	87	103	141
	PrecMax	2092	461	496	671	529	460	609	461	192	213	532	244	462
	PrecMin	753	21	48	37	26	16	7	2	1	0	9	25	22
	PrecStd	373	101	119	141	126	105	120	98	41	43	99	60	103
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	35.7	33.2	29.4	27.1	24.0	25.3	28.4	32.2	33.5	35.2	36.1
		MCWB	25.1	25.6	23.8	21.7	19.3	17.2	16.3	16.2	17.9	20.4	21.5	23.2
	2%	DB	33.3	32.5	30.2	27.4	25.0	22.2	23.0	25.5	28.6	30.7	32.2	33.1
		MCWB	24.0	24.5	23.0	20.6	18.0	16.0	14.8	15.5	17.8	19.9	21.5	22.9
	5%	DB	31.2	30.5	28.5	26.0	23.6	20.8	21.2	23.4	26.3	28.4	29.8	30.6
		MCWB	23.6	23.5	22.2	20.0	17.3	15.3	14.6	14.8	17.2	19.4	21.5	22.5
	10%	DB	29.5	28.8	27.1	24.7	22.0	19.4	19.5	21.4	24.4	26.2	27.8	28.8
		MCWB	23.3	22.6	21.5	19.6	16.6	14.9	13.6	14.0	16.7	18.9	20.6	22.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.8	24.8	23.2	21.4	19.2	18.9	19.3	20.8	23.6	24.3	25.7
		MCDB	33.2	33.3	30.3	27.1	24.4	21.7	22.7	25.4	28.1	30.0	31.6	33.0
	2%	WB	25.5	25.5	23.9	21.9	19.9	17.9	17.1	17.2	19.6	21.8	23.1	24.4
		MCDB	31.0	31.1	28.5	25.4	23.1	20.1	20.0	22.3	26.0	27.3	29.5	30.5
	5%	WB	24.6	24.6	23.2	21.1	18.6	16.8	15.8	16.1	18.6	20.7	22.2	23.5
		MCDB	29.4	29.1	27.0	24.7	22.0	19.4	19.0	20.9	24.0	25.6	27.8	28.8
	10%	WB	23.8	23.5	22.4	20.2	17.7	15.9	14.8	15.2	17.7	19.7	21.5	22.6
		MCDB	28.2	27.4	25.9	23.5	20.6	18.3	18.0	19.5	22.5	24.5	26.4	27.3
Mean Daily Temperature Range	5% DB	MDBR	10.2	9.8	9.6	10.0	12.2	11.1	13.0	14.7	14.2	12.6	11.7	10.8
		MCDBR	14.0	13.1	12.3	12.6	13.9	12.9	15.6	18.1	17.7	17.3	15.7	14.8
		MCWBR	6.1	5.8	5.6	6.4	7.5	7.3	8.4	8.9	8.1	7.6	6.7	6.2
	5% WB	MCDBR	11.8	11.8	10.3	10.9	11.4	9.5	11.4	14.3	14.5	13.5	12.9	12.2
	MCWBR	5.5	5.5	5.1	6.4	7.2	6.2	7.3	8.0	7.4	6.5	6.0	5.6	
Clear-Sky Solar Irradiance	taub	0.388	0.373	0.356	0.349	0.314	0.309	0.300	0.313	0.353	0.379	0.384	0.394	
	taud	2.488	2.545	2.587	2.570	2.638	2.642	2.651	2.594	2.476	2.433	2.464	2.456	
	Ebn at Noon	953	951	936	894	885	867	893	919	924	934	952	950	
	Edh at Noon	117	107	97	90	75	71	73	86	106	118	118	121	
All-Sky Solar Radiation	RadAvg	6.46	5.86	4.97	4.22	3.50	2.90	3.36	4.25	5.28	5.90	6.38	6.43	
	RadStd	0.53	0.48	0.28	0.20	0.22	0.19	0.27	0.33	0.42	0.53	0.54	0.65	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-58	+169	+104

Nomenclature: See separate page